HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

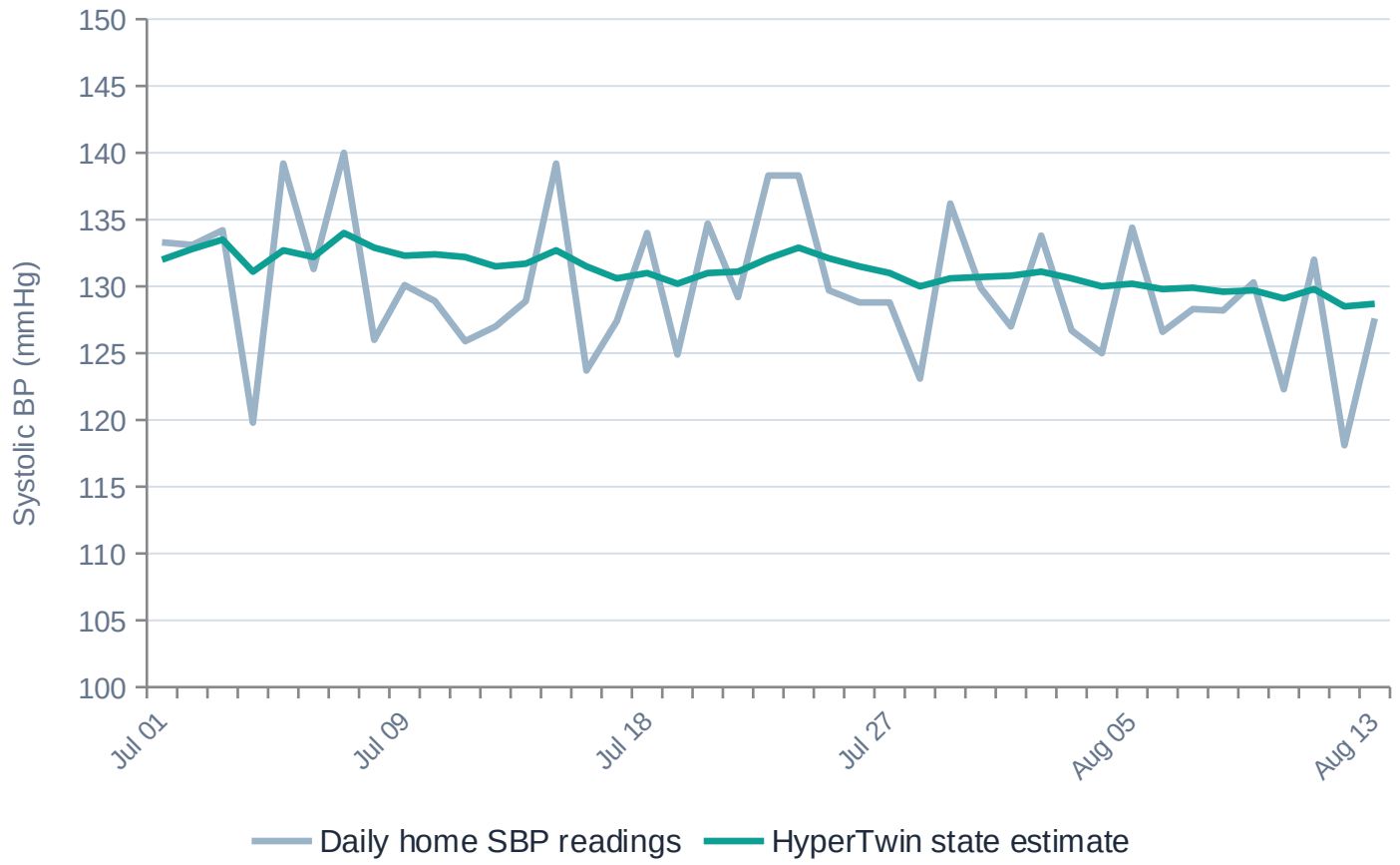
HyperTwin

From isolated home blood-pressure readings to a continuously updated, patient-specific state

Research prototype | synthetic data only | not a diagnostic or treatment tool

[Team name] · [College / incubator] · to be filled with verified details

One home reading is one noisy number



Synthetic patient P090, 41 valid morning readings

Home BP is guideline-endorsed

Out-of-office BP is recommended in major hypertension guidance.

Thresholds are for averages

Guidelines cite 135/85 mmHg for home BP averaged over several days.

The trend is what a clinician needs

Single readings scatter. A state estimate shows where the patient is heading.

Sources: ESH Working Group home BP position paper 2021 (PMC9904446); AAFP summary of ISH guidelines 2021

The insight

Read history and today's data together, as one evolving state.

Historical EHR

age, BMI, medications, family history, last clinic BP

+

New observations

home BP, sleep, steps, one day at a time

=

Patient state

a level and its uncertainty, updated every day

HyperTwin

HyperTwin is a research prototype that updates a patient-specific blood-pressure state from historical health information and home readings to forecast next-morning systolic BP risk.

Inputs

Synthetic EHR, home SBP/DBP, sleep, steps

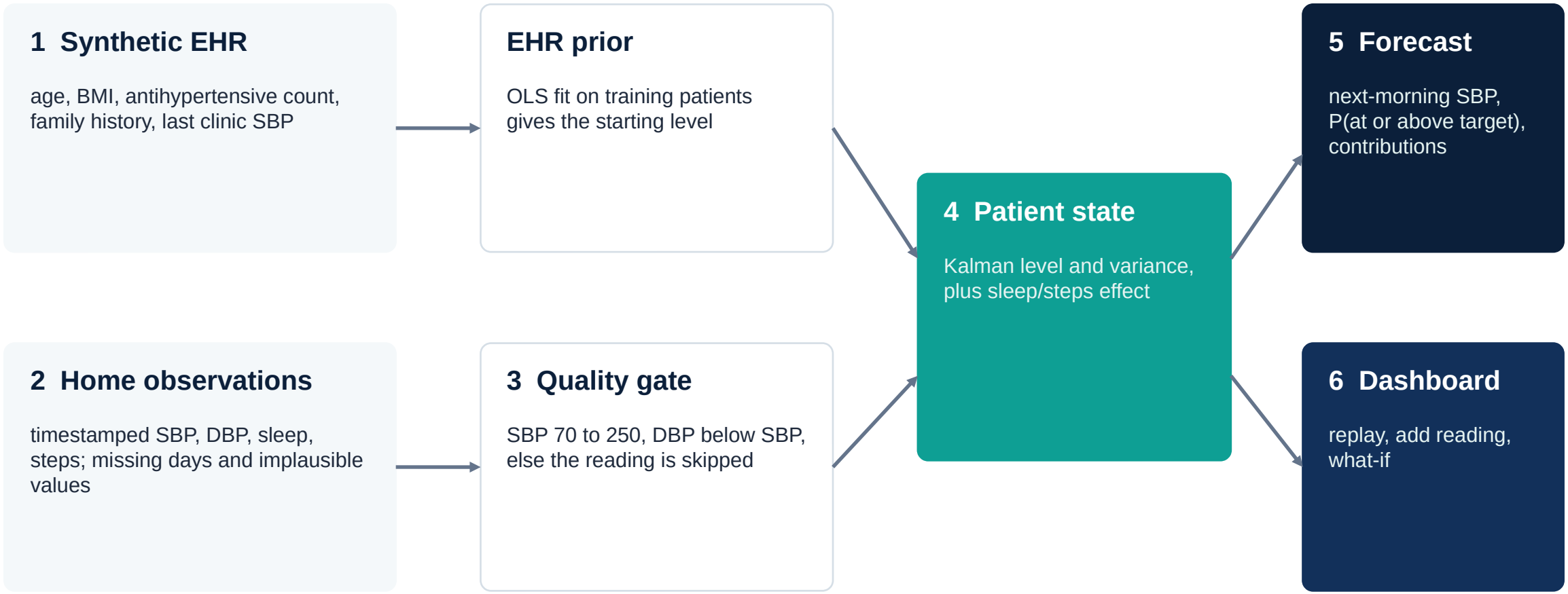
State

Kalman level and variance per patient

Outputs

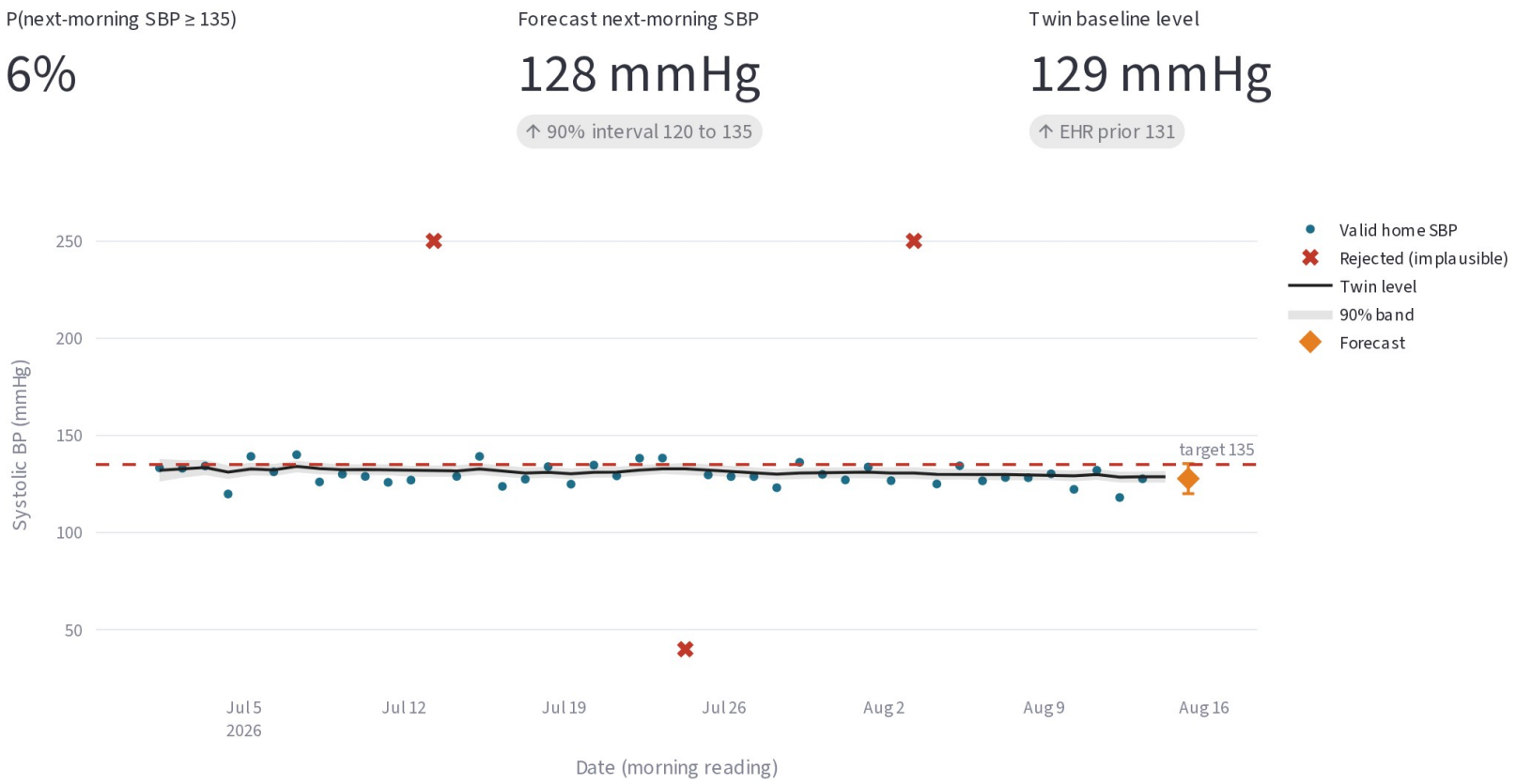
Next-morning SBP, P(SBP at or above target), contributions

How it works: data flow



Matches src/twin.py. Parameters fit on 90 training patients only; each forecast uses data before that day.

The clinician view



Forecast and risk

Next-morning SBP, 90% interval, P(at or above 135)

State over time

Twin level with 90% band against valid readings

Data quality

Rejected implausible readings shown as red crosses

Actual dashboard screenshot. Synthetic patient P090, replay day 45.

A new reading changes the state

Before

P(next-morning SBP ≥ 135)

6%

Forecast next-morning SBP

128 mmHg

↑ 90% interval 120 to 135

Twin baseline level

129 mmHg

↑ EHR prior 131

After adding 175/95 mmHg

P(next-morning SBP ≥ 135)

53%

↑ +47% after added obs

Forecast next-morning SBP

135 mmHg

↑ 90% interval 128 to 143

Twin baseline level

136 mmHg

↑ EHR prior 131

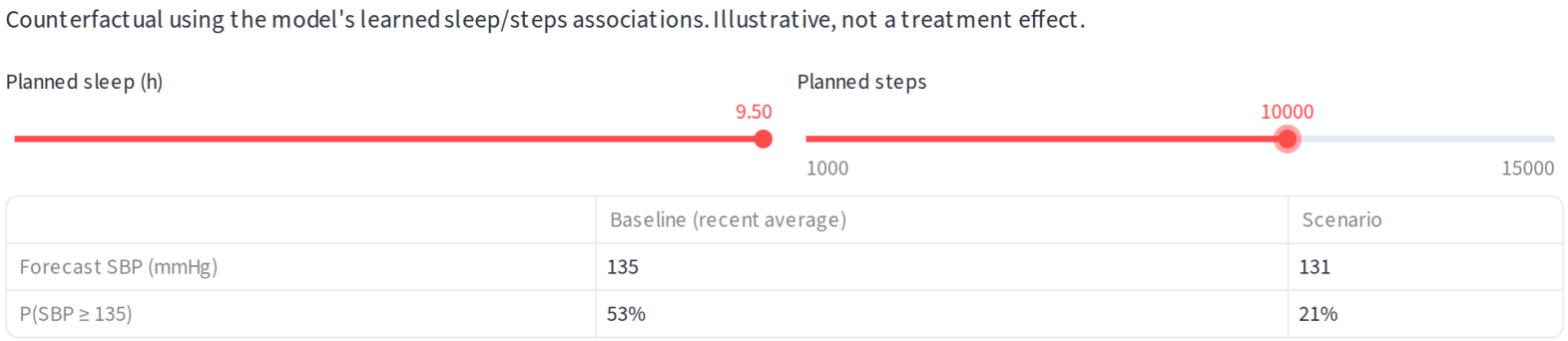
6% to **53%**

risk of next-morning SBP at or above 135 mmHg

A default 150/90 reading gives 6% to 20%.

Illustrative synthetic demonstration, patient P090 at replay day 45. Not a clinical outcome, and not what any real patient would see.

What if sleep and activity change?



	Baseline (recent average)	Scenario
Forecast SBP (mmHg)	135	131
P(SBP ≥ 135)	53%	21%

53% to 21%

forecast 135 to 131 mmHg, with 9.5 h sleep and 10,000 steps planned, after the 175/95 reading

How to read it

Uses the sleep and steps associations the model learned. In this prototype those effects are assumptions built into the synthetic data.

An illustration, not a treatment effect.

Actual dashboard screenshot, What-if scenario tab. Extreme settings chosen to make the effect visible.

Does the state beat simple baselines?

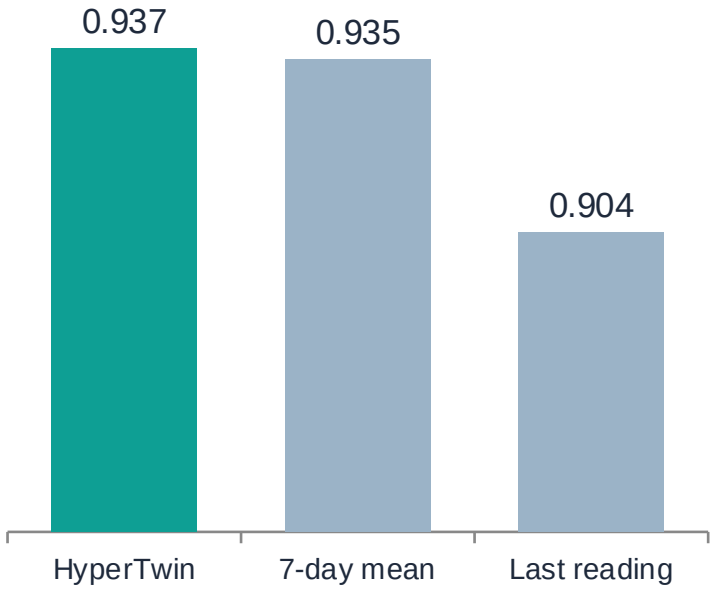
MAE (mmHg)

Lower is better



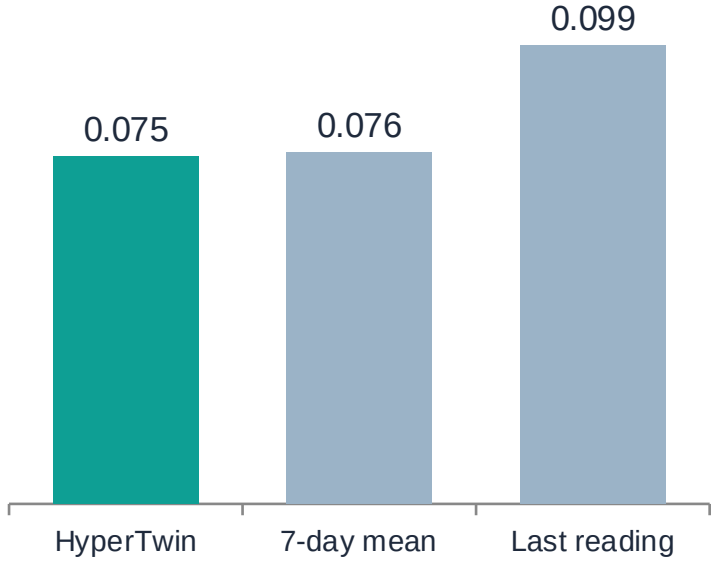
AUROC

Higher is better (axis starts at 0.85)



Brier score

Lower is better



Clear gain over the last reading. Small gain over the 7-day mean, not shown to be statistically significant.

Synthetic data. 4,024 predictions, 60 held-out test patients (90 training), one seed, no confidence intervals. Twin uses past data only.

What the prototype demonstrates

1

Patient-specific state

Starts from each patient's EHR, then learns from readings

2

Bad data handled

Missing and implausible readings do not move the state

3

Short-horizon forecast

Next-morning SBP with a 90% interval and risk

4

Interpretable output

Forecast split into EHR prior, recent level, sleep, steps

5

What-if simulation

Compare planned sleep and steps on the same patient

No first-of-its-kind claim. Many challenge entries build twins for other conditions.

Safety and limitations

Synthetic data only

No real patient information.

Not clinically validated

Results show agreement with our own generator.

Not a diagnosis or treatment tool

Research prototype.

Proxy target

Guidelines use multi-day averages. We predict a single morning.

Small edge over a 7-day mean

Not shown to be significant. One seed.

Assumed effects

Sleep and steps effects are set in the generator.

From readings to a state

A reading says what happened. A patient-specific state says where things are heading.

Validation roadmap

1. Test on licensed real home-BP data
2. Add confidence intervals and calibration checks
3. Forecast sleep and steps instead of assuming them
4. Review usefulness with clinicians

Team

[Team name]
[Member names]
[College / incubator]
[GitHub repository link]

Placeholders to be replaced with verified details before submission.